

Research on the Thrust Vector Control via Jet Vane in Rapid Turning of Vertical Launch

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Abstract—In order to solve the problem of rapid turning of missile vertical launch, the thrust vector control of jet vane is designed in this paper. By applying the force analysis to the jet vane, the mathematical model of the missile's pitch motion in the vertical plane is established. At the same time the jet vane controller is designed, which introducing the missile's pitch angle position control and rate control. The simulation results show that the jet vane controller with the PID control rule can make the missile fast turn within one second. The model of jet vane thrust vector control can effectively provide sufficient control torque for a missile's quick turn.

Keywords—jet vane; thrust vector control; vertical launch; digital simulation

I. INTRODUCTION

The incoming missiles such as cruise missiles may come from any direction, and are constructed to fly at low altitudes, just above treetops or water surfaces, avoiding enemy radar's detection. In such situations, a targeted ship may have just a few seconds identifying the threat and then to take countermeasures, such as firing a defensive missile. The vertical launch technology gives anti-aircraft missiles the ability to defend a full range of saturation attack. It has the advantages of quick response, targeting effective, efficient stowage, fast reloading, reliability and the protection of the weapon load. Every country over the world is now keeping an eye onto it. The missile using aerodynamic control plates processes a poor high-alpha pitch over ability in the launching stage as the initial velocity is low. The plates can't provide sufficient aerodynamic force to make a fast pitch over under that situation. The thrust vector control (TVC) system was developed as an alternative in order to resolve the problem. The TVC system can allow the missile to turn quickly and stably in any direction by deflecting the direction of the nozzle exhausts. TVC technologies can mainly be categorized as secondary injection of gas into the nozzle gas flow, the movable nozzle, the jet vanes, the jet tabs, and many others. The jet vane is the most widely used among them because of not only its simple structure but also the capability to do roll, pitch and yaw control. Figure 1 shows the comparison of the trajectories between jet vane and aerodynamic control.

In many previous studies, many aspects of the jet vane TVC system are seriously studied. These methods require the

detailed aerodynamic parameters of the jet vane, which requires a well design in advance. In this paper the design is aimed at a specific type of rocket. In the case of knowing the jet vane area, a mathematical model is established for the pitch motion of the missile flight in the vertical plane by applying force analysis to the vane. At the same time a jet vane servo controller is designed based on PID control technology. Through the software simulation, it can verify whether the preliminary design meets the requirements.

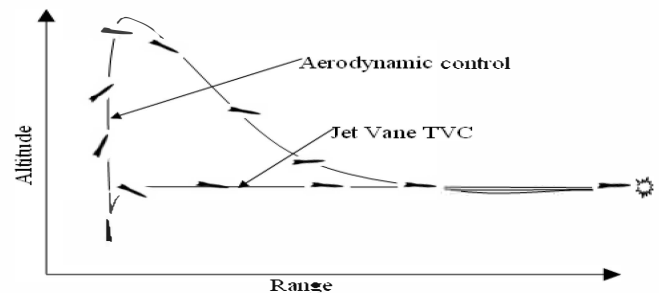


Fig. 1. Comparison of pitch turn between jet Vane TVC and aerodynamic control.

II. TURNING IN VERTICAL LAUNCHING

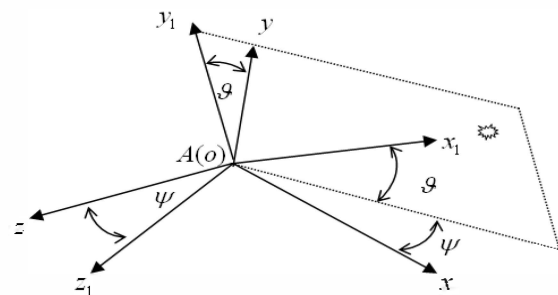


Fig. 2. Shooting plane (where $\gamma = 0$).

In vertical launching, the missile flight consists of three phases, the rapid turn, the midcourse and the terminal homing. In the first stage, a missile should quickly turn to intercept the target after hitting the safe altitude, in other words, adjust itself into the shooting plane. For the earth fixed inertial coordinate system (xyz) and the body coordinate system ($x_1y_1z_1$), during the

initial launch phase, the missile's attitude angles are roll (γ), pitch (θ), and yaw (ψ). The roll, and yaw attitudes are firstly adjusted through the missile controller, putting the missile and the target into the same longitudinal plane. Then pitch turnings are made within that plane. After the propellant burns out, the missile starts entering the midcourse phase. Shooting plane is shown in Figure 2.

In this design, the height of the launch tower is 6 meters, in additional with another safety height of 3 meters, the missile starts turning at 9 meters. As the initial velocity of missile is 30 m/s, according to the simple equation of motion, the calculated turn time is 0.3s. This design focuses mainly on verifying whether the jet vane can provide sufficient lateral control, therefore it is only to study the pitch control of the missile in the vertical plane.

III. MODELING OF THE VERTICAL LAUNCH TVC

A. Mechanical Analysis of the Jet Vane

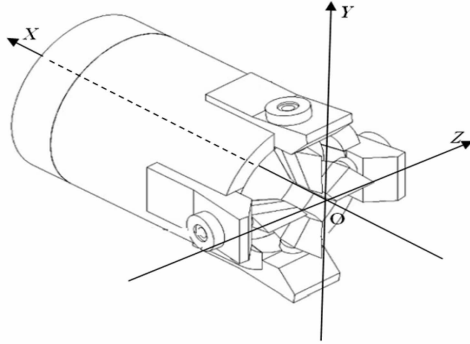


Fig. 3. The missile body coordinate system.

In the body coordinate system, the origin is assumed to be at the axial cross point of the two pairs of vanes, the X axis is the missile's longitudinal axis, forward to the warhead, and the Y axis is in the longitudinal plane of the missile body, perpendicular to the X axis and points up, the Z axis is perpendicular to the XY plane, and the direction is determined by the right hand coordinate system, as shown in Figure 3. In the jet vane coordinate system, the axis is parallel to the vane root chord, consistent with the gas flow direction, and it is the tangential force axis. The axis is perpendicular to the vane, and points up, being the normal force axis, the axis perpendicular to the plane, and parallel to the span wise. As shown in Figure 4.

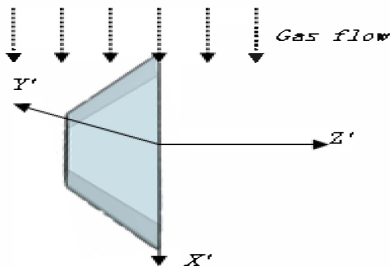


Fig. 4. The jet vane coordinate system.

Jet vane works in a high-speed and high-temperature gas flow. The basic working principal of a jet vane is the same to a supersonic wing, which is holding an angle of attack in the air free stream. As the jet vane deflects a certain angle relative to the gas flow, an oblique shock produces on a windward side of the jet vane which introduces the surface pressure increase, at the same time the expansion waves generates on a leeward side of the jet vane which causes the surface pressure to decrease. The pressure difference between the windward and leeward sides of the jet vane will produce a force onto the jet vane. The force can be divided into two components, one is normal, which controls the flight attitude of the missile body, the other is tangential, which can result in thrust loss. The purpose of the force analysis to the jet vane is to analyze the forces of the missile body. Therefore, as analyzing the force generated by the jet vane, its coordinate system coincides with the body's. It is shown in Figure 5.

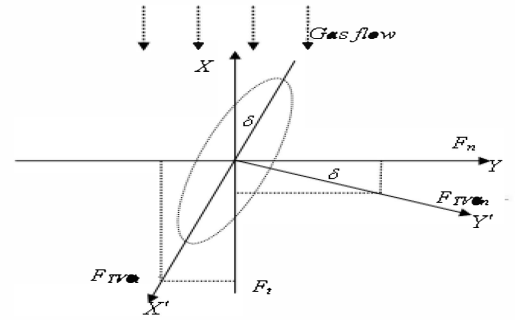


Fig. 5. Jet vane force analysis.

Figure 5 illustrates that the normal force (F_{TVcn}) and the tangential force (F_{TVct}) of the jet vane are resolved along the missile's body-axis system. X and Y are the missile body's coordinate system axes, and are the vane's body coordinate system axes. The chord line of the jet vane is the connection line between the leading edge and trailing edge, and parallel to the axis. Jet vane deflection angle is the angle between the gas flow and the chord line. If the chord line rotates counter-clockwise to be parallel to the gas flow line, the is positive, otherwise the is negative. In the missile body coordinate system, the normal force F_{TVcn} can be decomposed along the Y axis positive and along the X axis negative. Then the lateral force F_n and the resistance F_t working on the missile body can be calculated as follows:

$$F_n = F_{TVcn} \cos \delta - F_{TVct} \sin \delta \quad (1)$$

$$F_t = F_{TVcn} \sin \delta + F_{TVct} \cos \delta \quad (2)$$

According to the theory of aerodynamics and the supersonic linearized theory, the normal force f_{TVcn} and the tangential force f_{TVct} generated by one of the jet vanes can be calculated as follows:

$$f_{TVcn} = c_{TVcy} \frac{1}{2} \rho' V'^2 S_j = \frac{1}{2} \rho' V'^2 S_j \frac{4\delta/573}{\sqrt{M_a'^2 - 1}} \quad (3)$$

$$f_{n\alpha} = c_{n\alpha} \frac{1}{2} \rho V^2 S_j = \frac{2\rho V^2 S_j}{\sqrt{M_a^2 - 1}} \left[\left(\frac{\delta}{57.3} \right)^2 + \left(\frac{t_b}{B_d} \right)^2 \right] \quad (4)$$

where c_{TVcy} is the coefficient of lift and the c_{TVcx} is the coefficient of drag which both lie in the jet vane's axis system, the is the density of gas flow, is the velocity of gas flow, is the mach number of gas flow, is the reference area of jet vane, is the thickness of jet vane root chord and is the length of jet vane root chord.

As the jet vanes works in pairs, the normal force and tangential force generated by two jet vanes can be calculated as follows:

$$F_{TVcn} = 2f_{TVcn} \quad (5)$$

$$F_{TVct} = 2f_{TVct} \quad (6)$$

When δ is 0, it can be seen from formulas (1), (2), (3), (4), (5), (6) that the normal force F_{TVcn} on those vanes is 0, so does the lateral force of the missile body F_n . While the resistance F_{TVct} is generated by the thickness of the vane body, at the same time plays as a resistance on missile body.

B. Modeling of Missile Pitching Motion

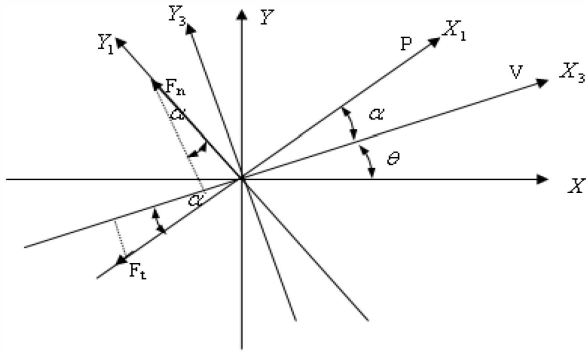


Fig. 6. The relationship between the speed coordinate system and the missile body coordinate system (sideslip angle $\beta = 0$).

The paper then adopted the coordinate transformation analysis to establish the equation of motion of the missile. Figure 6 illustrates the relative relationship among the missile body coordinate system, the speed coordinate system and the inertial coordinate system. As missile pitch turns in the vertical plane, the sideslip angle β is 0. In the Figure 6, the inertial coordinate system is denoted by (X, Y) , the missile body coordinate system is denoted by (X_1, Y_1) and the speed coordinate system is denoted by (X_3, Y_3) . The α is the angle of attack, and the θ is trajectory inclination. The lateral force F_n and the resistance F_t acting on the missile body can be converted to the speed coordinate system by coordinate transformation.

According to the missile aerodynamics theory, the dynamic equations of the missile motion can be written as follows:

$$m \frac{dV}{dt} = P \cos \alpha - mg \sin \theta - X - F_t \cos \alpha - F_n \sin \alpha \quad (7)$$

$$mV \frac{d\theta}{dt} = P \sin \alpha - mg \cos \alpha + Y - F_t \sin \alpha + F_n \cos \alpha \quad (8)$$

Many factors affect the missile's pitching moment in the vertical plane, but there are three main factors: the torque's static stability caused by attack angle, moment of pitch damping and the pitch control torque caused by jet vane deflection. The kinetic equation of missile around the centroid is shown as follows:

$$J_z \frac{d\omega_z}{dt} = M_z^\alpha \alpha + M_z^{\bar{\omega}_z} \bar{\omega}_z + F_n (x_c - x_{jc}) \quad (9)$$

where $\bar{\omega}_z$ is the angular velocity and its dimension is 1. It is calculated as follows:

$$\bar{\omega}_z = \frac{\omega_z L}{V} \quad (10)$$

The aerodynamic force and the aerodynamic torque expression are given by the following expressions:

$$X = c_x \frac{1}{2} \rho V^2 S \quad (11)$$

$$Y = c_y \frac{1}{2} \rho V^2 S \quad (12)$$

$$M_z^\alpha \alpha + M_z^{\bar{\omega}_z} \bar{\omega}_z = (m_z^\alpha \alpha + m_z^{\bar{\omega}_z} \bar{\omega}_z) \frac{1}{2} \rho V^2 SL \quad (13)$$

If the missile's center of gravity changes, the missile's static steady moment coefficient's correction formula is shown as follows:

$$m_z^\alpha \alpha = m_{z0}^\alpha \alpha + c_y (Xg - Xg_0) / L \quad (14)$$

According to the missile's flight principles and its kinematic equations in the vertical plane, the mass change equation and the positional relationship can be written as follows:

$$\frac{d\theta}{dt} = \omega_z \quad (15)$$

$$\frac{dx}{dt} = V \cos \theta \quad (16)$$

$$\frac{dy}{dt} = V \sin \theta \quad (17)$$

$$\frac{dm}{dt} = -m_c \quad (18)$$

$$\alpha = \theta - \theta \quad (19)$$

C. The Design of the Jet Vane Controller

The alternation of a missile's speed vector relies mainly on changing the combined external forces acting on the missile. The external forces acting on the missile include total aerodynamic force (R), thrust (P) and gravity (G). In the initial turn boost phase, the missile's speed is relatively low and the role of aerodynamic is very limited. At the same time the gravity is only affected by the mass of the solid rocket engine, while its size rarely changes and the direction is always pointing to the center of the earth. Therefore, in order to achieve a fast turn, changing the thrust vector is quite necessary while changing the vane's deflection angel can alter the rocket's size and the direction of thrust.

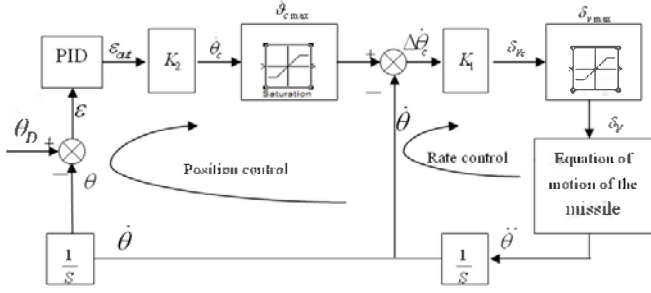


Fig. 7. Jet vane controller block diagram.

The jet vane controller mainly contains with the missile's pitch angle position control and rate control, as shown in Figure 7. In the position control loop, in order to reduce and eliminate the deviation quickly between the actual and desired pitch angle of the missile, the PID control algorithm is added. It can also reduce the oscillation. In accordance with the design of the controller, the control law is shown as follows:

$$\varepsilon = \vartheta_D - \vartheta \quad (20)$$

$$\varepsilon_{out} = K_p \left\{ \varepsilon(n) + \frac{T}{T_i} \sum_{i=0}^n \varepsilon(i) + \frac{T_D}{T} [\varepsilon(n) - \varepsilon(n-1)] \right\} \quad (21)$$

$$\dot{\vartheta}_c = k_2 \varepsilon_{out} \quad (22)$$

$$\delta_c = k_1 (\dot{\vartheta}_c - \dot{\vartheta}) \quad (23)$$

where $\dot{\vartheta}_c$ is the pitch control rate, and its unit is degree/sec. ϑ_D is the desired pitch angle, ϑ is the actually pitch angle, δ_c is command deflection angle of the jet vane, these angle units are degrees. k_1 is the angular rate gain, with the unit of degree/degree/sec. k_2 is position gain, coming in degree/s/degree.

In the jet vane control system, the vanes and air rudders share one servo control system, because the aerodynamic control surface has deflection limit. That's why a deflection limit of the jet vane δ_{vmax} should be set. At the same time in order to ensure the pitch rate complies with the tolerant torque acting on the jet vane, a pitch rate control limit is set as $\dot{\vartheta}_{cmax}$

IV. SIMULATION DATA ANALYSIS

In the design, the initial parameters of the missile are as follows: its initial velocity V_0 is 30 meters/s, the initial pitch angle is 90° , the initial trajectory inclination is 90° , the thrust of the rocket engine is $15G$, the limit of jet vane deflection angle is 25° , the pitch rate control limit is $250^\circ/s$, start turning time is $0.3s$, the expected pitch angle at the end of turning is 0° and the working time of the booster rocket engine is $5s$.

According to the control principle of the jet vane controller, the command of turning is carried out at $0.3s$, and the deviation between the initial pitch angle and the desired pitch angle reaches its maximum, and that deviation instantly makes the pitch angle control rate to reach a saturation value which is 250 degrees/s, as shown in Figure 8. At the same time, the jet vane's deflection angle has reached the saturation value of 25° , and maintains approximately $0.1s$, as shown in Figure 9. During the process of the turning, the missile's pitch angle from the vertical state can quickly pitch to 0° within $1s$. As the pitch angle closes to the expected 0° , the torque generated by the jet vane is only used to balance the aerodynamic torque which is caused by the angle of attack, pitch rate and the alternation in attack angle's changing rate. The pitch angle remains around 0° for some time, so that the trajectory angle which is lag catches up with the angle of attack, until the angle of attack is zero. As shown in Figure 10.

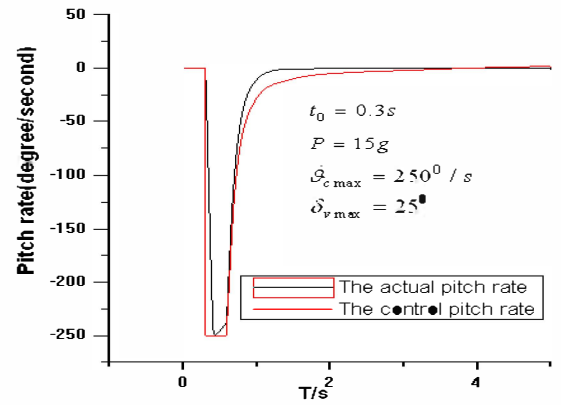


Fig. 8. The pitch rate curve.

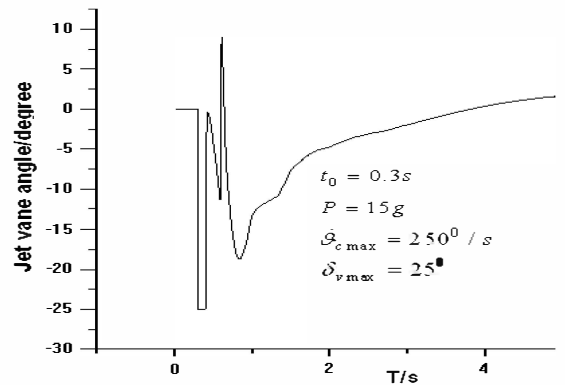


Fig. 9. Jet vane deflection angle curve.

At the same time it can be seen from figure 8 and 9 that when the actual pitch rate of the missile is close to the control command of the pitch rate, the jet vane starts to change the deflection direction in order to prevent the missile to turn over and leave the required control direction. It can be known from the Figure 11 in which the ballistic curve is shown that before the turning time of 0.3s, the missile rises vertically, and at the 0.3s the speed is 74.8 meters/sec with a height of 15.9 meters, completely beyond the safe altitude. When the booster rocket engine stopped working, the speed of missile is 703 meters/s, with a height of 164.5 meters and a range of 1542.3 meters. Then it can begin the proportional navigation guidance to intercept short-range and low-altitude incoming targets.

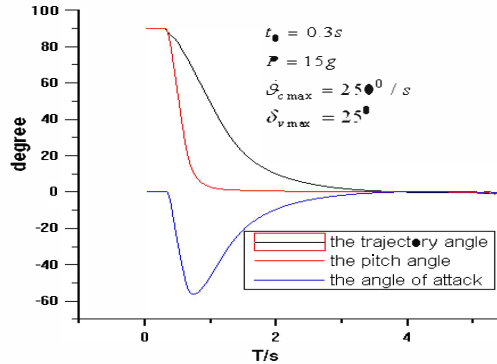


Fig. 10. The pitch angle, angle of attack and trajectory angle curve.

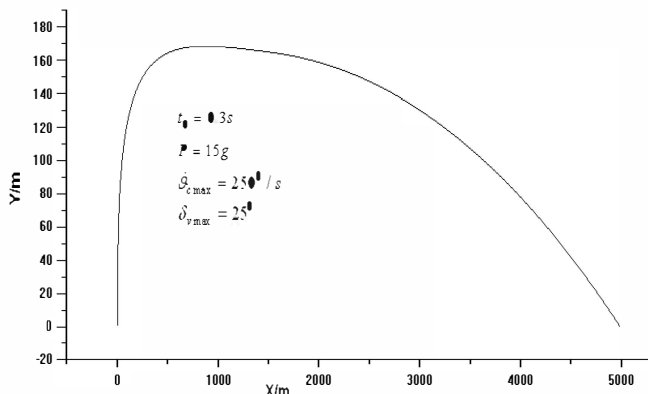


Fig. 11. Trajectory curve.

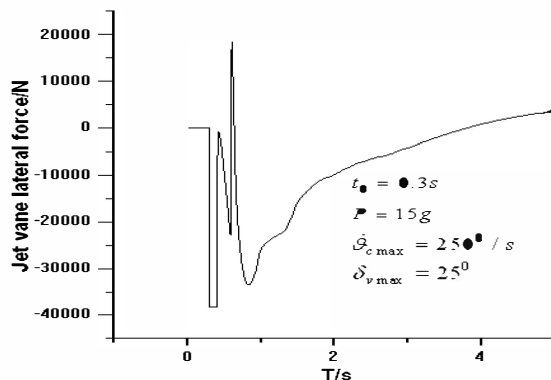


Fig. 12. Jet vane lateral force curve.

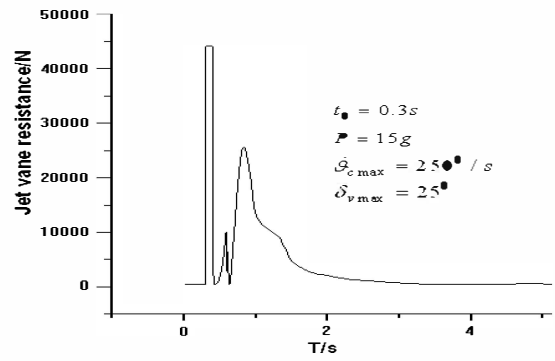


Fig. 13. Jet vane resistance curve.

Figure 12 and 13 illustrate the curve of the normal force and the resistance of the jet vane versus time. If the deflection of angle of the jet vane is at its maximum, the normal force is the largest and so does the resistance of the vane. That's why the jet vane will cause great thrust loss.

V. CONCLUSION

In this paper the jet vane thrust vector control technology is the key of the missile fast turning control system. It is not only a simple structure with a good accessibility, but also being able to provide sufficient control torque for a missile's quick turn. Results from the simulation show that the research methods, the model of the missile's body motion and the jet vane controller can be applied to the engineering applications.

The inadequacies of this design include those as unable to analyze the hinge moment generated by the jet vane. The hinge moment determines the power of the servomotor of the jet vane, and should be as small as possible according to the analysis from the aspect of controls.

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